

# LING 420: Computational Linguistics and Neural NLP

Term: Fall 2026 | Instructor: Dr. Noam Chomsky (nchomsky@ling.univ.edu)

## Course Description & Overview:

Theoretical foundations and deep learning models for syntax, vector semantics, Transformer attention architectures, and LLMs.

## COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

| Assignment Description                          | Weight | Due Module | Deliverable Format               |
|-------------------------------------------------|--------|------------|----------------------------------|
| N-Gram Language Model & POS Tagger              | 25%    | Week 04    | Python Model & Accuracy Report   |
| Transformer Self-Attention Decoder from Scratch | 35%    | Week 08    | PyTorch Implementation & Weights |
| Multilingual Information Extraction Project     | 30%    | Week 11    | NER Pipeline & Empirical Paper   |
| NLP Research Paper Presentation                 | 10%    | Week 06    | ACL Paper Summary Deck           |

## WEEKLY TERM SCHEDULE & READING REQUIREMENTS

| Timeline | Technical Topic Focus                               | Assigned Readings & Docs                    |
|----------|-----------------------------------------------------|---------------------------------------------|
| Week 01  | Module 01: Tokenization & Byte-Pair Encoding        | Jurafsky & Martin (Ch. 2)                   |
| Week 02  | Module 02: N-gram Models & Smoothing Techniques     | Jurafsky & Martin (Ch. 3)                   |
| Week 03  | Module 03: Vector Semantics & Word2Vec              | Mikolov et al. (2013)                       |
| Week 04  | Module 04: Sequence Labeling with HMMs              | Rabiner (1989)                              |
| Week 05  | Module 05: Recurrent Neural Networks & LSTMs        | Hochreiter & Schmidhuber (1997)             |
| Week 06  | Module 06: Transformer Architecture & Attention     | Vaswani et al. (2017)                       |
| Week 07  | Module 07: Masked Language Modeling: BERT           | Devlin et al. (2018); Liu et al. (2019)     |
| Week 08  | Module 08: Autoregressive LLMs & Instruction Tuning | Radford et al. (2019); Ouyang et al. (2022) |
| Week 09  | Module 09: Syntax: CFGs & Dependency Parsing        | Chomsky (1957); Chen & Manning (2014)       |
| Week 10  | Module 10: Semantic Parsing & QA Systems            | Berant et al. (2013); SQuAD Documentation   |
| Week 11  | Module 11: Final Project Showcase & Ethics in NLP   | Bender et al. (2021)                        |